

Rectangular Government Survey System

The **rectangular government survey system** -also known as the Public Land Survey System (PLSS), because it was originally used to survey federally owned blocks of land-is regulated by the Bureau of Land Management under the U.S. Department of the Interior.

The rectangular government survey system is used in 30 western and southern states. It isn't used by East Coast states except Florida, and it also isn't used by Texas. The PLSS is actually a series of surveys that divide land into townships that are six miles by six miles square (for a total area of 36 square miles or 23,040 acres). Townships are further subdivided into 36 one-mile-square (640 acres) sections (each section is one square mile or 640 acres), then into quarter sections, quarter-quarter sections, or irregular government lots.

Beginning at an initial point, townships are surveyed in all directions. The north-south line that runs through the initial point is a true meridian (called the principal meridian). There are 37 principal meridians in the national survey, which are referred to when distinguishing among the various surveys. Surveyors also establish an east-west line from which all measurements originate, called a base line. The east-west lines of a survey are called township lines. The north-south lines are called range lines. As an example, a township might be identified as Township 6, Range 1 West, which would mean that it was the 6th tier of townships north of a base line and in the first column of townships west of a principal meridian.

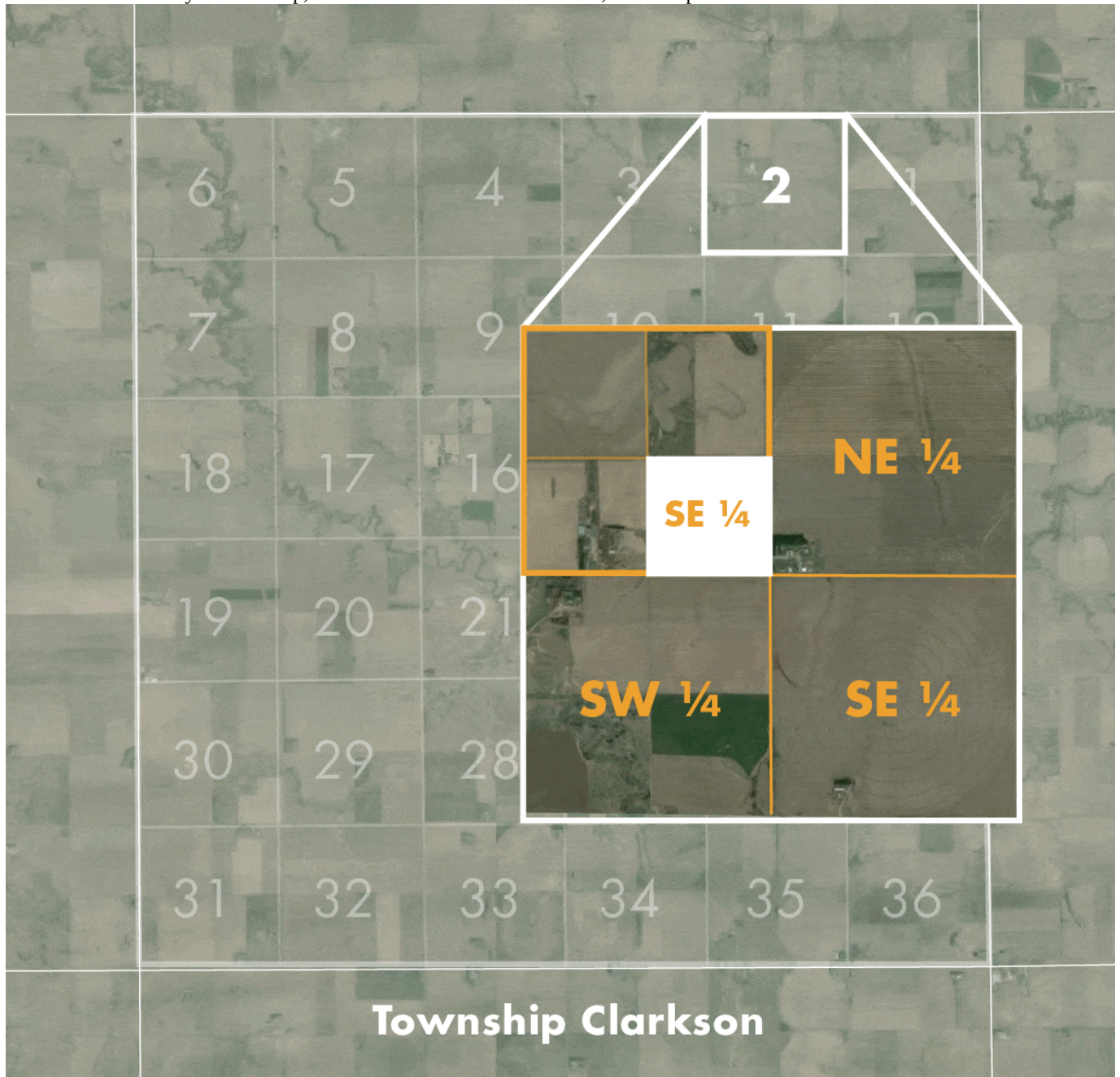
Here's a simple layout showing how a township is designated. Note that the numbers don't read left to right, but start at the northeastern corner and follow a zigzag pattern. You may see an exam question or two about this, so be sure you understand how the grid is laid out.



Legal descriptions using the PLSS system include the state, principal meridian name, township, and range designations with directions and the section number.

How to Read Rectangular Survey Descriptions

Look at this figure for the fictional township of Clarkson. This is described in "rectangular speak" as: $S\frac{1}{2}$ SE $\frac{1}{4}$ NW $\frac{1}{4}$ Section 2. It is further described by its township, such as: $S\frac{1}{2}$ SE $\frac{1}{4}$ NW $\frac{1}{4}$ Section 2, Township Clarkson.



When land is mapped on the rectangular survey system, locate the parcel by reading the directions from right to left. Therefore, in Township Clarkson, Section 2, we look for the northwest, and from that we find the southeast 1/4, and from there we find the south 1/2, which is the shaded portion.

How large is this parcel?

Let's look at the description: $S\frac{1}{2}$ SE $\frac{1}{4}$ NW $\frac{1}{4}$ Section 2, Township Clarkson. That tells us that the parcel is $\frac{1}{32}$ ($\frac{1}{2} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{32}$). Keeping in mind that the entire section is 640 acres, divide the 640 acres by 32 ($\frac{1}{32} \times 640 = 20$) to get 20 acres. The parcel is 20 acres.